

Solving multi-step equations



1

a 54

b 54

c Yes

2

a -0.176

b -1.19

c No, $n = 0.8$ is not a solution of the equation.

3

a No

b No

c No

d No

e Yes

4 $2(x - 12) + 9 = 3x - 8$

5 $80x + \frac{1}{6} = 10(8x + 2)$

6

a $k = 2$

b $r = 4$

c $r = 6$

d $r = 6$

e $k = 2$

f $v = 11$

g $k = 8$

h $n = -\frac{7}{3}$

i $y = -26.5$

7

a $x = 3$

b $x = -25$

c $x = 22$

d $x = 4$

e $x = 10$

f $x = 14$

g $x = 21$

h $x = 10$

i $x = -38.7$

j $x = \frac{64}{3}$

k $x = \frac{19}{20}$

l $x = \frac{15}{7}$

m $x = \frac{28}{3}$

n $x = \frac{1}{6}$

o $x = 6$

p $x = 2$

8

a $x = -2$

b $x = -6$

c $x = -13$

d $x = 5$

e $x = 3$

f $x = -0.2$

g $x = -3$

h $x = -5$

i $x = -2$

j $x = \frac{11}{35}$

k $x = \frac{5}{17}$

9

a No solutions

b One solution



c No solutions

d Infinitely many

e One solution

f No solutions

g One solution

h No solutions

i One solution

j No solutions

k Infinitely many

l Infinitely many

m No solutions

n One solution

o Infinitely many

p Infinitely many

10

a $x = -7$

b One solution

11

a

i $3n + 3 = 4n + 6$

ii One solution

b

i $2n + 6 = 2n + 5$

ii No solution

c

i $4n + 2 = 4n + 2$

ii Infinitely many solutions